

Technical Report Instructions

The typical flow of a technical article is: abstract, introduction, methodology, analysis, discussion/results, conclusion/limitations, appendix. It does NOT need to follow this exact format (some journal articles have slightly different formats).

Depending on how much you want to elaborate, you can split certain sections into two parts (i.e. Analysis could become Quantitative Analysis and Qualitative analysis.) You have free reign to decide your outline insofar as it makes logical sense.

Journal articles are often written for an audience that knows statistics well. That is, you will not need to elaborate terms such as p-values, null & alternative hypotheses, etc. It will be the responsibility of the reader to understand the statistical analysis method. However, it would be nice to have some plain language conclusions at the end for readability.

We do not expect students to write or perform literature review, ethical statements, conflicts of interest, or funding. (However, it would be nice to include literature review if you had the time.)

We are expecting students to use either LaTeX, R-Markdown, or Quarto for the report.

WARNING: DO NOT SAY AFFECT AND OTHER WORDS THAT IMPLY CAUSALITY. (SUCH AS: Influence, impact, contribute...)

This is not a controlled randomized trial! You should look for words such as association, relationship, connection.

1. Abstract

In prior versions of this course, we did not request for an abstract. However, abstracts are necessary for any academic article. Your abstract should be a quick summary of your paper and is shown before the introduction. They are used for other researchers to tell whether the paper is worth reading. It should address the following:

- A brief introduction to the topic.
- Aim of the paper.
- A brief statement regarding the data collection methodology.
- A summary of key findings.
- A brief overview of implications of results, or what needs to be improved.

Here's an example of an abstract (colour coded to match the above description):

In a statistics program, most courses emphasize statistical theory over practical applications, often resulting in a focus on examinations rather than assignments. However, many statistics programs include a course on survey and sampling design, where students can be assessed through projects. Designing these projects requires more resources than tests, including increased grading workload and time spent resolving group conflicts. This study examines whether students prefer projects over traditional examinations and identifies the benefits they perceive from project-based learning. In a third-year statistics survey & sampling course, we asked students their perceptions of project based learning through Google Forms after completion of the final project. The results show that 81% of students prefer projects and found them useful for developing skill sets necessary for the workforce. The main reasons for disliking projects were group conflicts and unclear instructions. In response, we plan to provide more resources to support student success and to find ways to mitigate potential group disputes in the future.

2. Introduction

Within the introduction, the following questions should be answered:

- What is your study about and why should we care?
- What background information is necessary to let the reader understand?
- What are your research questions?
- What are your hypotheses?
- What is the brief outline of the rest of your paper?

Below, a student wrote two different drafts of an introduction regarding a study including cannabis usage:

Draft 1:

Those opposed to the decriminalization of cannabis will often cite that it is destroying the youth and their cognitive ability to function in society. Cannabis is a popular drug that people occasionally will smoke for leisure activity. There are studies done that show that prolonged cannabis consumption is not good for brain development, and perhaps this could influence students to care less about school. We aim to analyze the following research question:

- (RQ1) What are the impacts of cannabis usage on lecture attendance?
 - Null hypothesis: cannabis usage has no association on lecture attendance.
 - Alternative hypothesis: cannabis usage is correlated with a lower attendance rate.

- (RQ2) What are the impacts of cannabis usage on students' grades?
 - Null hypothesis: cannabis usage has no association on students' grades.
 - Alternative hypothesis: cannabis usage is correlated with lower grades.
- (RQ3) Do students perceive cannabis to be a positive asset in their life?
 - Null hypothesis: cannabis usage is not associated with being an advantage to one's life.
 - Alternative hypothesis: cannabis usage is linked to improving one's lifestyle.

There are some issues with this introduction:

1. Fairly bland, boring, and sometimes awkward.
2. It doesn't transition well, i.e., the introduction of research questions were quite abrupt.
3. What is their exact population? Where could they be gathering this information?
4. What is the outline for the rest of the paper?

Draft 2:

This study examines the impact of cannabis consumption on university students' lecture attendance. Cannabis, a psychoactive substance frequently used by young adults for recreational purposes, has garnered increasing attention due to its potential implications for cognitive and behavioral outcomes, particularly among students. Extensive research has suggested that prolonged and frequent cannabis consumption may have adverse effects on brain development and cognitive function (Iversen, 2003).

In light of this, we formulate the hypothesis that individuals who engage in regular, weekly cannabis use are more prone to reduced lecture attendance. This research endeavors to investigate the relationship between cannabis consumption patterns and student engagement with academic activities, shedding light on an area of growing concern in contemporary education.

The structure of the paper is as follows: Section 2 outlines our data collection methods. Section 3 presents our quantitative analysis. In Section 4, we explore feedback from students and the common themes that arise. Section 5 discusses the qualitative results and addresses our research question. Section 6 covers the limitations of our study, and Section 7 concludes our analysis.

Again, there are some issues with this introduction:

1. This is not concise (mostly the second paragraph), and honestly boring to read. It sounds like there is an abundance of "fluff" to over-compensate for lack of substance. (What is contemporary education?)

2. The research questions and hypotheses are not clear. (They also don't mention enough.)
3. Again, we are unsure of the exact population and where they could possibly be gathering this information.

We can combine drafts 1 and 2, and integrate the missing pieces, to create a superior introduction (note: longer doesn't always mean better, it's just that the previous two drafts had missing information).

Final Draft:

Cannabis faces significant stigma from older generations due to its illegalization and negative stereotypes, such as the belief that it increases laziness. Moreover, extensive research suggests that prolonged and frequent cannabis consumption may have adverse effects on brain development and cognitive function (Iversen, 2003). While some individuals may use cannabis for recreational purposes, it can also provide relief from medical conditions that induce high levels of pain.

In this study, we analyze whether the stigma against cannabis is well-deserved. Specifically, we examine the impact of cannabis consumption on university students' lecture attendance and grades at a North American intensive university. In October 2024, we deployed a survey via email to collect data on students' cannabis usage, academic performance, demographic factors, and attitudes towards cannabis. The survey included both users and non-users of cannabis for comparisons. Additionally, we carefully differentiated between prescribed and recreational cannabis use. We aim to study the following research questions:

- (RQ1) What are the impacts of cannabis usage on lecture attendance?
 - Null hypothesis: cannabis usage has no association on lecture attendance.
 - Alternative hypothesis: cannabis usage is correlated with a lower attendance rate.
- (RQ2) What are the impacts of cannabis usage on students' grades?
 - Null hypothesis: cannabis usage has no association on students' grades.
 - Alternative hypothesis: cannabis usage is correlated with lower grades.
- (RQ3) Do students perceive cannabis to be a positive asset in their life?
 - Null hypothesis: cannabis usage is not associated with being an advantage to one's life.
 - Alternative hypothesis: cannabis usage is linked to improving one's lifestyle.

The structure of the paper is as follows: Section 2 outlines our data collection methods. Section 3 presents our quantitative analysis. In Section 4, we explore feedback from students and the common themes that arise. Section 5 discusses the

qualitative results and addresses our research question. Section 6 covers the limitations of our study, and Section 7 concludes our analysis.

3. Methodology

This is where you outline your data collection methodology in detail.

- **Where** and **when** was the data collected? (Piazza, lectures, tutorials, online databases...)
- What sampling method did you use? (SRS, stratified, etc....)
 - How did you **ensure** randomness? (randomly sampling from an R program, systematically deploying surveys in lecture based on seating arrangements...)
- What were your strata, if any?
- What was your sample size?
- What is a general summary of the questions you asked? (What are your variables?)

Below we present an example.

Between May 2023 to August 2023, a survey meant to understand students' academic performance and recreational drug usage was deployed within an introductory statistics course at a research-intensive North American university. We utilized simple random sampling by using a Python random generator to randomly sample 50 students from an email list of all students taking the introductory statistics course ($N = 242$). Out of 50 students, 19 did not respond, thus we were left with only ($n = 31$) responses. The survey consisted of 10 short answer questions, asking for their average cannabis usage per week, lecture attendance, and miscellaneous demographic factors (gender, program of study, age.)

4. Analysis

WARNING: In a journal article, codes are not provided unless in an appendix. Do not put R-code here.

Things that are included in this section:

- Show relevant graphs & tables.
- Show computations for your sample size.
- Necessary assumptions for tests are shown before utilizing them. (If assumptions are not satisfied, you should not be using that test!)
 - For example, the assumptions for the two-sample test for mean are the following:

- The samples are independent from each other & are obtained randomly.
 - The samples are normally distributed.
 - The variances for the two independent groups are equal.
- Show the outputs of the computations and statistical tests, i.e., p-values, test-statistics are provided here.
 - **Remark: the tests for significance should be done at the 0.05 level.**
- Some statistical information that may be useful to reference (depending on the context): standard deviation, confidence interval, median, IQR.

If you have used a questionnaire, use $N = 200$ for computing the sample size. n , unfortunately, is already known. So, you will be reverse calculating as if you wanted a certain bound to fit your sample size.

You do **NOT** need to show **EVERYTHING** you computed. Only the computations that you will **DISCUSS** in the report. Normally, a lot of side-things you compute at first will have no statistically significant results. (I.e., gender may have no connection with the amount of cannabis a person will smoke.)

Part of this project is having the ability to decipher what is important to report.

This is a snippet of what to include in the analysis section:

For our study, our population size is $N = 300$ and we plan to collect data using simple random sampling. To determine a sample size to collect, we'll go with the calculation focusing on the mean population parameter. It is assumed that there is an equal proportion amongst those who enjoy projects in courses versus those who enjoy examinations. Hence, given a bound of error of 0.13, the sample size calculation ends up being like:

$$\frac{Npq}{(N-1)D + pq} = \frac{(300)(0.5)(0.5)}{(299)\frac{0.13^2}{4} + (0.5)(0.5)} = 49.56 \approx 50$$

Hence, we sampled 50 students for our analysis.

We found that 40% of our participants identified as male ($n = 20$), and the rest were female ($n = 30$). We also had 64% domestic students ($n = 32$) and 36% international students ($n = 18$). Surprisingly, only 40% said they preferred projects over examinations ($n = 20$).

To answer RQ1, we need to calculate the one-sample proportion test. All of our assumptions are satisfied, as we have a random sample, a binomial distribution (two binary outcomes/choices) and we have at least 5 from each outcome ($np = 20$, $n(1-p) = 30$). We obtain a Chi-square test statistic of 2, $df = 1$, and $p = 0.1573$.

RQ2 tries to determine if certain demographic factors contribute to whether someone prefers assignments. Again, we need to check if the assumptions for the two-sample proportion test are satisfied. We have the same binomial distribution, and the samples are still independent from each other.

Below is a table that summarizes the last assumptions. All assumptions were satisfied, so we included a brief result of the statistical test.

Gender Assumptions				Domestic vs International Assumptions			
n1	p1	n2	p2	n1	p1	n2	p2
30	43.33333%	20	0.35	32	0.25	18	66.66667%
n1*p1	n1*(1-p1)	n2*p2	n1*(1-p2)	n1*p1	n1*(1-p1)	n2*p2	n1*(1-p2)
13	17	7	13	8	24	12	6
Test Statistic	DF	P-value	Significant?	Test Statistic	DF	P-value	Significant?
0.086806	1	0.7683	No	0.032552	1	0.8568	No

We also analyzed the Likert scale data, which had options which ranged from: “strongly disagree”, “disagree”, “slightly disagree”, “neutral”, “slightly agree”, “agree”, and “strongly agree”. We denoted “strongly disagree” as 1, “strongly agree” as 7.

To see whether students perceived their skills to be enhanced through project courses in general, we conducted the one-sample t-test for the mean, where option 4 (neutral) denotes no change. The assumptions for the one-sample test involves the data being randomly selected, independent from each other, and the normal appears normally distributed. The first two conditions have been addressed, and the last condition can be satisfied by the central limit theorem due to having 50 samples which is greater than 30.

Below is a summary of the Likert scale questions, as well as the one-sample test for the mean:

Category	Mean	SD	Median	IQR	Test Statistic	DF	p-value
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Collaboration	4.1	1.55511	4	2	0.4547	49	0.6513
Writing	4.54	1.95051	5	3	1.9576	49	0.05598
Interesting	3.72	1.552352	4	1.75	-1.2754	49	0.2082
Understanding	4.24	1.635823	4	2	1.0374	49	0.3046
Coding	4.26	1.827343	4	2.75	1.0061	49	0.3193

Remark 1: some interesting data points to include would be the confidence interval, which I omitted due to laziness. I also did not include “advanced tests”.

Remark 2: it was unrelated to any of the research questions, but if you peek at the attached code file I tested to see if there were any differences amongst the perceptions of Likert scale answers between male & female, and international & domestic. There were no differences, so I did not include them. If you have time for more results, and they do end up being significant, it would be interesting to include them.

5. Discussion/Results

In this section you should write the interpretation of your results, and hopefully in plain language conclusions.

You may also use this section to elaborate facts regarding a table or a graph in detail, such as pointing out abnormalities or highlighting the stark differences between two groups in your data.

A discussion section is typically longer the more interesting the results, but it can be made relatively short if no results were found.

6. Limitations

Briefly mention limitations and what should be done in the future. Consider what you would’ve changed in your survey if you went back in time to redo the study. A common limitation in studies involve an inadequate sample size, failing to reject the null hypothesis, missing confounding variables, biased sample size and/or survey, poor survey questions.

A common issue in prior years is that students mentioned limitations that were more of a critique of the method as opposed to their own study. They probably stole these

ideas from ChatGPT. As a warning, the following are not sufficient limitations (and I will address why):

- Lack of causal connections. STA304's official name is "Survey, Sampling, & Observational Data". Observational data, in nature, will never lead to causal results. If you want to make a causal connection, you're in the wrong course.
- Self-reported data. You want to stalk and hack into a database that reveals information about your fellow classmates just for your project!? The course name also includes "surveys"! Surveys will always be tied with self-reported data.
- Temporal factors. Maybe this would cause an effect... But realistically, are you going to spend a year collecting survey data? I doubt most people's opinions don't drastically change unless your project is entirely season dependent...

There are more nonsensical answers that generative AI tools will spout out. I am begging you to take a second to think about your limitations rather than consulting AI. (In fact, I may have literally given the answers in the first paragraph of this section.)

Here's an example of an adequate limitation section:

In our study we tried to see which demographic factors (gender, ethnicity, domestic or international, and year of study) are linked with an overreliance of using AI tools using the Chi Square test for independence or the non-parametric Fisher's exact test (depending on whether the assumptions of the Chi square were satisfied).

Unfortunately, in all cases we failed to reject the null hypothesis. Hence, there are no ties between various demographic factors and whether students tend to use AI tools. In fact, we found that 90% of students admitted to using AI generative tools. As a result, we are less likely to see any relationships due to a high majority already using them. In the future, we will incorporate open-ended questions to ask students to elaborate on their preferences.

7. Conclusion

In this section you should have the following:

- Summarize your findings and explicitly answer your research questions.
 - No new information should be provided; everything mentioned here should have been mentioned in the past.
- Talk about more in detail of what researchers should do in the future.

Naturally, people may wonder the difference between an abstract and the conclusion.

- The abstract is supposed to be a concise summary of the entire paper. It goes through the phrases of an introduction, methodology, analysis, and results.

- The conclusion does summarize the paper but emphasizes on the latter half of the paper (results and limitations). It does not include background information. It will also re-address limitations and go into depth about future directions.

8. Appendix

If you have references, include them first. Then, include your R-code on a new page. Your R-code must be clean and presentable, and it will be checked whether the R-code works. Other users should be able to run your code without issue. The purpose is to allow other people to repeat your computations to double verify. R is **mandatory** for this project. **Using other programming languages or software will result in a 0 for the coding portion, even if it's for a partial element of the project.**

Below is an example of clean R code. (Note: such data does not exist.) Within the R-code, include comments for the user to read and understand what the lines are for.

```
# Loading relevant libraries
library(tidyverse)

# Changing working directory - needs to change depending on the
machine
setwd("C:/Users/AnnaH/OneDrive/CannabisData")

# Loading the data
data = read.csv("STA304_DataProgressReportExample.csv")

# Introductory data analysis
mean(data$lecture_attendance)
mean(data$cannabis_usage)
sd(data$lecture_attendance)
sd(data$cannabis_usage)

attach(data)

# Looking at the linear regression model
modell = lm(lecture_attendance ~ cannabis_usage, data = data)

# Creating a linear regression plot
plot(lecture_attendance, cannabis_usage,
     main = "Lecture Attendance verses Cannabis Usage",
     ylab = "Average Cannabis Usage (per week)",
     xlab = "Lecture Attendance rate (whole semester)")
```

Technical Report (Rough Draft) Rubric

- Completed abstract. (/0.25)
- Completed introduction & methodology. (/0.25)
- Completed analysis & discussion, with no R codes included. (/0.25)
- Completed limitations & conclusion (and FOLLOWED the instructions for the limitation section.) (/0.25)
- Completed appendix. (All code is COMPLETED in R.) (/0.25)
- Included various drafts, one before generative AI tools and one after. Also included ChatGPT dialogues (could be from a link). (/0.25)
 - If no one used generative AI tools, then specify in the rough draft somewhere that no one used AI.
 - Remark: we have caught students in the past for using generative AI tools due to wording and nonsensical logic. Lying about not using ChatGPT can constitute an academic offense.
- The report was written in LaTeX, R Markdown, or Quarto. (/0.25)
 - Appropriate files (.tex, .rmd, etc.) must be attached.
- The report is formatted, and has satisfactory spelling & grammar. (/0.25)

Technical Report (Final Draft) Rubric

Format (/1)

- The report was written in LaTeX, R Markdown, or Quarto. (/0.8)
 - Appropriate files (.tex, .rmd, etc.) must be attached. Omitting such files will result in a 0 for this section.
- Satisfactory spelling & grammar throughout the entire report. (/0.1)
- The report flows through a logical flow. (/0.1)

Abstract (/1)

- Briefly introduces the topic. (/0.2)
- Briefly discusses the aim. (/0.2)
- Briefly discusses the data collection methodology. (/0.2)
- Summarizes key findings. (/0.2)
- Discusses implications of results. (/0.2)

Introduction (/0.5)

- Explains the background information in sufficient detail for the audience to understand what the study is about. (/0.2)
- Provides research questions and hypotheses. (/0.2)
- Provides a brief outline of the rest of the paper. (/0.1)

Methodology (/0.5)

- Explains where and when the data was collected. (/0.1)
- Explains the sampling method, including strata & clusters (if any), and how randomness was ensured. (/0.2)
- The population (the group of interest) should be obvious. Provides the sample size and a general summary of questions asked. (/0.2)

Analysis (/3.5)

- Summary statistical information is provided, such as proportions, means, standard deviations, confidence intervals, median, IQR, etc. (/0.25)
- Relevant graphs & tables are given. (/0.5)
- The graphs & tables are formatted and labelled appropriately. (/0.25)
- Provided correct computations for sample sizes. Additionally, sample sizes for certain sub-categories are provided. (/0.25)
 - For example, if you are comparing male vs female, list the number of male participants and female participants.
- All necessary assumptions for tests are shown/addressed before utilizing them. (/0.75)
- The simple statistical test(s) are performed correctly, with relevant outputs provided (such as test statistic, df, p-values.) (/0.75)
- The advanced statistical test(s) are performed correctly, with relevant outputs provided. (/0.75)

Discussion (/1)

- All research questions are answered with sufficient evidence from the analysis section. (/0.5)
- Provided correct interpretations of the analysis, as well as plain language conclusions. Does NOT provide causal interpretations. (/0.5)

Limitations (/0.5)

- At least two reasonable limitations are addressed. (/0.25)
- Discusses ways to mitigate such limitations in the future and other ways to improve the study. (/0.25)

Conclusion (/0.5)

- Sufficiently summarized the results in the study with correct plain language conclusions. (/0.25)
- Discusses implications of the results and future directions in detail. (/0.25)

Appendix (/0.5)

- All relevant R codes are shown. (/0.25)

- The code is accompanied by appropriate comments. (/0.25)

Implementing Feedback (/1)

- Students considered the instructor's feedback of their rough draft and implemented all changes.
- No part marks will be implemented; failure to incorporate all feedback will result in a zero for this section.

Marks can be further deducted for the following reasons:

- **Email submission (-20%)**
- **Plagiarism of any examples on this page and past reports (-20%)**
- **Using programming languages to create graphs other than R. I.e., using Excel, Python, etc. (-20%)**